

STATS 503, Homework 5

Please use a Jupyter notebook to write up your solutions. Submit your work through Canvas by uploading an html file that contains the html export of your notebook file. The html file should include all your code (and therefore no hidden cells). Use markdown formatting to indicate which part of your notebook corresponds to each problem.

1. [3 POINTS] ISLP, chapter 5, conceptual exercise #2, parts a, b, c, d, e, f
2. [3 POINTS] ISLP, chapter 5, conceptual exercise #4
3. In this problem, we will perform cross-validation on a simulated data set.

(a) [2 POINTS] Generate a simulated data set using the following code:

```
rng = np.random.default_rng(1)
x = rng.normal(size=100)
y = x - 2 * x**2 + rng.normal(size=100)
```

In this data set, what is n and what is p ? Write out the model used to generate the data in equation form.

- (b) [1 POINT] Create a scatterplot of X against Y . Comment on what you find.
- (c) [4 POINTS] Construct 5 folds using `sklearn.model_selection.KFold`, and specify `random_state=3`. Using these folds, compute the cross-validation errors that result from fitting the following four models using least squares. **Use the same five folds for all four models.**
 - i. $Y = \beta_0 + \beta_1 X + \varepsilon$
 - ii. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$
 - iii. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$
 - iv. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \varepsilon$.

For each fold and each model, report the mean-squared error. Which of the models had the smallest average error? Is this what you expected? Explain your answer.

4. [2 POINTS] Consider estimating log odds using logistic regression with ridge penalties, setting the regularization strength via LOOCV. True or false: the result will depend on a *random seed* that we use as part of the LOOCV process. Explain your answer.
5. [8 POINTS] ISLP, chapter 5, applied exercise #9. This problem relates to the Boston dataset. If you like, you can load this dataset by installing the ISLP Python package. Alternatively, you can also download it from our course Canvas site.

6. In this problem, we will compute the LOOCV error for a logistic regression model on the `Weekly` data set, which can be downloaded from Canvas.
- (a) **[1 POINT]** Fit a logistic regression model that predicts `Direction` using `Lag1` and `Lag2`. Report and comment on the result.
 - (b) **[1 POINT]** Fit a logistic regression model that predicts `Direction` using `Lag1` and `Lag2` using all but the first observation. Report and comment on the result.
 - (c) **[1 POINT]** Use the model from (b) to predict the direction of the first observation. Was this observation correctly classified?
 - (d) **[3 POINTS]** Write a for loop from $i = 1$ to $i = n$, where n is the number of observations in the data set, that performs each of the following steps:
 - i. Fit a logistic regression model using all but the i th observation to predict `Direction` using `Lag1` and `Lag2`.
 - ii. Use this model to predict the direction for the i th observation.
 - iii. Determine whether or not an error was made in predicting the direction for the i th observation. If an error was made, then indicate this as a 1, and otherwise indicate it as a 0.
 - (e) **[1 POINT]** Take the average of the n numbers obtained in part (d)(iii) of this problem in order to obtain the LOOCV estimate for the test error. Comment on the results.

Total points for this assignment: 30